

GENERALIZED LELONG NUMBERS DETERMINE VALUATIVE TRANSFORMS

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ABSTRACT. Boucksom, Favre, and Jonsson proved that equality of the valutive transforms of two plurisubharmonic germs implies equality of all generalized Lelong numbers with tame reference weights. We prove the converse. It is enough to test against the analytic weights defined by primary ideals. The proof first polarizes the resulting diagonal intersection identities and then applies a strict relative Hodge index statement on each birational model.

1. INTRODUCTION

Let o denote the origin of $\mathbb{C}^n$, and let $u, v : (\mathbb{C}^n, o) \rightarrow \mathbb{R} \cup \{-\infty\}$ be plurisubharmonic (psh) germs, neither identically equal to $-\infty$. If φ is a psh weight with an isolated pole at o , Demailly's generalized Lelong number is

$$(1) \quad \nu_\varphi(u) = (dd^c u \wedge (dd^c \varphi)^{n-1})(\{o\}).$$

The definition, comparison theorem, and invariance under bounded changes of the weight are established in [Dem87].

We use the terminology of [BFJ08, Section 5.3]. A *psh weight* is a psh germ φ with an isolated singularity at o such that e^φ is continuous. It is *tame* if there is a constant $C > 0$ such that, for every $t > 0$ and every $f \in \mathcal{I}(t\varphi)$,

$$(2) \quad \log |f| \leq (t - C)\varphi + O(1).$$

Because φ is negative near its pole, inequality (2) says that the singularities detected by the multiplier ideals approximate φ with an error uniformly bounded in t . We shall not use the definition directly; it is included to make clear which class of test weights occurs in the theorem.

Boucksom–Favre–Jonsson associate to u a valutive transform $\hat{u}$. The same information can be written as a nef Weil b -divisor $Z(u)$ over o : the coefficient of $Z(u)$ along an exceptional prime divisor is the negative of the corresponding generic Lelong number. We recall the construction in

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section 2. The language of b -divisors goes back to Shokurov; the local form used here is developed in [BFJ08, Sections 1–4]; see also [Sho03].

The following four conditions appear in [BFJ08, Theorem A]:

- (1) u and v have the same Lelong number at every infinitely near point;
- (2) $\mathcal{J}_+(tu) = \mathcal{J}_+(tv)$ for every $t > 0$;
- (3) the relative types of u and v with respect to every tame maximal psh weight agree;
- (4) $\nu_\varphi(u) = \nu_\varphi(v)$ for every tame psh weight φ .

Here $\mathcal{J}_+(w)$ is the stationary value of $\mathcal{J}((1+\varepsilon)w)$ as $\varepsilon \downarrow 0$. Boucksom–Favre–Jonsson proved that (1), (2), and (3) are equivalent and imply (4). They left (4) $\Rightarrow$ (1) open and proposed the construction of fundamental solutions for the formal Monge–Ampère operator as one possible approach; see [BFJ08, p. 455 and p. 457].

In dimension two, the valuative transform and its Laplacian are governed by the valuative tree; see [FJ05]. In higher dimension, the formal Monge–Ampère operator is instead built from intersections on all modifications. The strong openness theorem [GZ15] settles the openness problem mentioned alongside Theorem A in [BFJ08], but it does not invert the averaging operation in Theorem B of that paper. More recent intersection theories for nef b -divisor classes establish general approximation and Hodge index principles in a projective setting; see [DF22]. The local argument below uses only the intersection formalism already available in [BFJ08], but we make every extension to signed Cartier tests explicit.

We prove this remaining implication. The statement below is slightly stronger: only weights with analytic singularities coming from primary ideals are needed.

Theorem 1.1. *Let $u, v : (\mathbb{C}^n, o) \rightarrow \mathbb{R} \cup \{-\infty\}$ be psh germs. The following conditions are equivalent, provided neither germ is identically $-\infty$.*

- (i) u and v have the same Lelong number at every infinitely near point.
- (ii) $\nu_\varphi(u) = \nu_\varphi(v)$ for every tame psh weight φ .
- (iii) For every $\mathfrak{m}_o$ -primary ideal $\mathfrak{a} = (f_1, \dots, f_N) \subset \mathcal{O}_{\mathbb{C}^n, o}$,

$$\nu_{\varphi_{\mathfrak{a}}}(u) = \nu_{\varphi_{\mathfrak{a}}}(v), \quad \varphi_{\mathfrak{a}} := \frac{1}{2} \log \left(\sum_{j=1}^N |f_j|^2 \right).$$

Consequently, all four conditions of [BFJ08, Theorem A] are equivalent.

Here is the proof in outline. Put $W = Z(u) - Z(v)$. For a primary ideal $\mathfrak{a}$, the analytic weight $\varphi_{\mathfrak{a}}$ satisfies

$$(3) \quad \nu_{\varphi_{\mathfrak{a}}}(u) = -\langle Z(u), Z(\mathfrak{a})^{n-1} \rangle.$$

Thus condition (iii) gives diagonal equations

$$(4) \quad \langle W, Z(\mathfrak{a})^{n-1} \rangle = 0.$$

Products of ideals turn sums of their divisors into further test divisors. Expanding (4) for these products recovers every mixed intersection. The ideal divisors span the Cartier b -divisors. On a fixed model X_π , we may therefore test W against its own incarnation W_π and against a relatively ample exceptional class H . This gives

$$W_\pi^2 \cdot H^{n-2} = 0.$$

A strict relative Hodge index theorem forces $W_\pi = 0$. Since the model was arbitrary, $W = 0$.

Each input is separated below. Sections 2 and 3 review the b -divisor and intersection formalism, section 4 proves the exact strict Hodge statement required, and section 5 writes out polarization before the proof of the theorem.

2. BIRATIONAL MODELS AND VALUATIVE DIVISORS

2.1. Models above the origin. We work with a cofinal system of smooth projective modifications

$$\pi : X_\pi \longrightarrow (\mathbb{C}^n, o)$$

that are isomorphisms away from o . Replacing a modification by a smooth dominating one does not lose any divisorial valuation. Resolution of singularities and principalization provide such dominating models. The local Riemann–Zariski space is constructed in [BFJ08, Sections 1.1–1.2].

Write $\text{Exc}(\pi)$ for the exceptional locus and set

$$(5) \quad \text{Div}(\pi) := \bigoplus_{E \subset \text{Exc}(\pi)} \mathbb{R}E.$$

Every exceptional prime is proper over o . Since X_π is smooth, an exceptional divisor is determined by its relative numerical class, and

$$(6) \quad \text{Div}(\pi) \simeq N^1(X_\pi/\mathbb{C}^n)_\mathbb{R};$$

see [BFJ08, Section 1.2]. If $\mu : X_{\pi'} \rightarrow X_\pi$ is a higher model, exceptional divisors admit pushforward and pullback, with $\mu_*\mu^*D = D$.

We use $\mathbb{R}$ -Cartier b -divisors throughout, and usually omit the prefix “ $\mathbb{R}$ ”. The following elementary consequence of projectivity will be used twice.

Lemma 2.1 (An ample exceptional divisor). *For every smooth projective modification $\pi : X_\pi \rightarrow (\mathbb{C}^n, o)$ there is an integral exceptional divisor A such that $\mathcal{O}_{X_\pi}(A)$ is π -ample. After replacing A by a positive multiple and shrinking the representative of the base germ, it may be taken π -very ample.*

Proof. Projectivity provides a relatively ample numerical class. The isomorphism (6) identifies the relative Néron–Severi space with the space of exceptional real divisors. Since the ample cone is open, it contains a rational exceptional class. After multiplication this class is represented by an integral exceptional divisor A . Relative ampleness depends only on the numerical class, and a sufficiently large positive multiple of a relatively ample

line bundle is relatively very ample. These are the relative forms of the standard ampleness criteria; see [Laz04, Chapter I, Section 1.7]. $\square$

Definition 2.2. A *Weil b -divisor* Z over o is a compatible collection $Z = (Z_\pi)_\pi$ with $Z_\pi \in \text{Div}(\pi)$ and $\mu_* Z_{\pi'} = Z_\pi$ whenever π' dominates π . The divisor Z_π is the *incarnation* or *trace* of Z on X_π .

A *Cartier b -divisor* C is a Weil b -divisor for which some model X_π satisfies $C_{\pi'} = \mu^* C_\pi$ on every higher model. Such a model is a *determination* of C .

Thus $D \in \text{Div}(\pi)$ defines a Cartier b -divisor by pullback to higher models; denote it by $\overline{D}$ when needed. A Weil b -divisor need not be determined on any finite model. These are the projective- and inductive-limit definitions in [BFJ08, Section 1.2]. Two Weil b -divisors are equal precisely when their incarnations agree on every model.

For later use, we also recall positivity. A Cartier b -divisor C determined on X_π is *nef over o* if

$$(7) \quad C_\pi \cdot \Gamma \geq 0$$

for every complete curve $\Gamma \subset \pi^{-1}(o)$. This criterion is [BFJ08, Proposition 2.2]. A nef Weil b -divisor is, by definition, a coefficientwise limit of nef Cartier b -divisors. Thus “nef” here is a relative condition over the germ, not nefness on a compact ambient variety.

2.2. The divisor attached to a psh germ. Let E be an exceptional prime on X_π . At a very general point of E , $u \circ \pi$ has a generic Lelong number $\nu_E(u)$. Define

$$(8) \quad \text{ord}_E Z(u) := -\nu_E(u).$$

Equivalently, if

$$dd^c(u \circ \pi) = \sum_E \nu_E(u)[E] + R_\pi$$

is the Siu decomposition and R_π does not charge an exceptional prime, then $[Z(u)_\pi] = -\sum_E \nu_E(u)[E]$. This is the construction in [BFJ08, Section 5.2].

Proposition 2.3 (Boucksom–Favre–Jonsson). *For every psh germ u , the Weil b -divisor $Z(u)$ is nef.*

Proof. This is [BFJ08, Theorem 5.1]. The approximation behind it is useful to recall. The ideal divisors $k^{-1}Z(\mathcal{J}(ku))$ are nef Weil b -divisors (not necessarily Cartier) and converge coefficientwise to $Z(u)$. Formula (5.3) of [BFJ08] bounds $Z(u)$ between the divisor of its multiplier ideal and that divisor minus the log-discrepancy function. Division by k makes the error tend to zero. Nef Weil b -divisors are defined as the closure of the nef Cartier cone; see [BFJ08, Sections 2.1 and 3.4]. $\square$

Remark 2.4 (Signs). The coefficients in (8) are nonpositive. This is the convention of [BFJ08]. For example, on the blowup of o , $-E$ is relatively ample and has positive degree on curves contracted by the blowup.

2.3. Infinitely near points. Let $p \in \pi^{-1}(o)$ and let $\mu : \text{Bl}_p X_\pi \rightarrow X_\pi$ be the blowup of the smooth point p , with exceptional divisor F_p . In local coordinates centered at p , the order along F_p is the ordinary multiplicity at p . Its psh counterpart is

$$(9) \quad \nu_{F_p}(u) = \nu^L(u \circ \pi, p),$$

where ν^L is the usual Lelong number. This follows directly from the definition after pullback by the ordinary blowup and is the observation used in [BFJ08, Section 5.6]. Consequently,

$$(10) \quad Z(u) = Z(v) \iff \nu^L(u \circ \pi, p) = \nu^L(v \circ \pi, p) \text{ for all } (\pi, p).$$

For the forward direction, compare coefficients on the blowup at p . For the reverse direction, fix an exceptional prime E and choose a very general smooth point $p \in E$ outside the other exceptional components. The ordinary Lelong number of $u \circ \pi$ at such a point is the generic Lelong number $\nu_E(u)$; this is the generic-value property in the Siu decomposition used in [BFJ08, Section 5.2]. Equality at all infinitely near points therefore gives equality of the coefficient of every exceptional prime in (8).

3. PRIMARY IDEALS, ANALYTIC WEIGHTS, AND INTERSECTIONS

3.1. The analytic and formal local rings. The first four sections of [BFJ08] are written for an algebraic local ring and its completion, whereas our ideals lie in the analytic local ring $\mathcal{O}_{\mathbb{C}^n, o}$. This causes no ambiguity for primary ideals. Indeed, if $\mathfrak{a}$ is $\mathfrak{m}_o$ -primary, then $\mathfrak{m}_o^N \subset \mathfrak{a}$ for some N . Its integral-closure and divisorial data are therefore determined by a finite jet, and extension to the completion followed by contraction preserves $\mathfrak{a}$ and all values $\text{ord}_E(\mathfrak{a})$. Conversely, polynomial primary ideals are analytic primary ideals. This is the identification used at the beginning of [BFJ08, Section 5]. In particular, [BFJ08, Proposition 1.14] and the intersection results of its Section 4 apply to the ideal divisors considered below.

3.2. The Cartier b -divisor of an ideal. Let $\mathfrak{a} \subset \mathcal{O}_{\mathbb{C}^n, o}$ be $\mathfrak{m}_o$ -primary. For each exceptional prime E , set

$$(11) \quad \text{ord}_E Z(\mathfrak{a}) := -\text{ord}_E(\mathfrak{a}), \quad \text{ord}_E(\mathfrak{a}) := \min_{f \in \mathfrak{a}} \text{ord}_E(f).$$

On a log resolution π of $\mathfrak{a}$,

$$(12) \quad \mathfrak{a} \mathcal{O}_{X_\pi} = \mathcal{O}_{X_\pi}(Z(\mathfrak{a})_\pi).$$

On higher models this divisor pulls back. Thus $Z(\mathfrak{a})$ is Cartier; see [BFJ08, Section 1.2]. For blowups and their tautological line bundles, see [Har77, Chapter II, Section 7]. The sheaf in (12) is generated relative to the base, so $Z(\mathfrak{a})$ is nef; compare [BFJ08, Proposition 2.2].

Valuation additivity gives, for primary ideals $\mathfrak{a}, \mathfrak{b}$,

$$(13) \quad Z(\mathfrak{a}\mathfrak{b}) = Z(\mathfrak{a}) + Z(\mathfrak{b}),$$

and more generally

$$(14) \quad Z(\mathfrak{a}_1^{k_1} \cdots \mathfrak{a}_r^{k_r}) = \sum_{i=1}^r k_i Z(\mathfrak{a}_i).$$

These are equations (2.1) of [BFJ08]. A product of $\mathfrak{m}_o$ -primary ideals remains primary because its radical is $\mathfrak{m}_o$.

Proposition 3.1 (Ideal divisors span). *The real vector space of Cartier b -divisors is spanned by the divisors $Z(\mathfrak{a})$ of $\mathfrak{m}_o$ -primary ideals. Thus every Cartier b -divisor has an expression*

$$(15) \quad C = \sum_{j=1}^N \lambda_j Z(\mathfrak{a}_j), \quad \lambda_j \in \mathbb{R}.$$

Proof. This is the real form of [BFJ08, Proposition 1.14]. We recall its geometric content. If an integral Cartier b -divisor is determined by an integral exceptional divisor D on X_π , choose A as in [theorem 2.1](#). Since A is nonzero and nef, [BFJ08, Proposition 2.4] shows that every exceptional coefficient of A is negative. For $m \gg 1$, the divisors mA and $D + mA$ therefore have non-positive coefficients, and both $\mathcal{O}_{X_\pi}(mA)$ and $\mathcal{O}_{X_\pi}(D + mA)$ are generated relative to the base. Put $D_1 = D + mA$, $D_2 = mA$, and

$$\mathfrak{a}_i := \pi_* \mathcal{O}_{X_\pi}(D_i) \subset \mathcal{O}_{\mathbb{C}^n, o}.$$

Because D_i is exceptional and nonpositive, $\mathfrak{a}_i$ is an ideal whose zero set is supported at o ; after increasing m it is $\mathfrak{m}_o$ -primary. Relative generation gives a surjection $\pi^* \mathfrak{a}_i \twoheadrightarrow \mathcal{O}_{X_\pi}(D_i)$, hence

$$\mathfrak{a}_i \mathcal{O}_{X_\pi} = \mathcal{O}_{X_\pi}(D_i).$$

Consequently $D_i = Z(\mathfrak{a}_i)_\pi$, and the associated Cartier b -divisor is $Z(\mathfrak{a}_i)$. Since $D = D_1 - D_2$, integral Cartier b -divisors are generated as an abelian group by ideal divisors. Tensoring with $\mathbb{R}$ gives (15). $\square$

Remark 3.2. The coefficients λ_j may be negative. Thus [theorem 3.1](#) concerns the full vector space, not only the nef cone. This permits the later test against the signed Cartier divisor $\overline{W}_\pi$.

3.3. The analytic weight of an ideal. For generators $\mathfrak{a} = (f_1, \dots, f_N)$, set

$$(16) \quad \varphi_{\mathfrak{a}} = \frac{1}{2} \log \left(\sum_{j=1}^N |f_j|^2 \right).$$

Two generating sets change this weight by a bounded function: each generator of either set is a bounded holomorphic linear combination of the other set. Demailly's comparison theorem implies that the generalized Lelong number is unchanged; see [Dem87].

Since $\mathfrak{a}$ is primary, its zero set is $\{o\}$. Moreover $e^{\varphi_{\mathfrak{a}}} = (\sum_j |f_j|^2)^{1/2}$ is locally Lipschitz. Lemma 5.10 of [BFJ08] says that a psh germ with Hölder-continuous exponential is tame. Hence $\varphi_{\mathfrak{a}}$ is a tame psh weight. Its generic Lelong number along E is $\text{ord}_E(\mathfrak{a})$, so

$$(17) \quad Z(\varphi_{\mathfrak{a}}) = Z(\mathfrak{a}).$$

3.4. Intersection products. For $D_1, \dots, D_n \in \text{Div}(\pi)$, their local intersection number $D_1 \cdots D_n$ is computed on X_π . It is defined because the divisors are supported over o , so the resulting zero-cycle is proper. The product is symmetric, multilinear, and preserved on higher models by the projection formula; see [BFJ08, Section 4.1, Proposition 4.2] and [Ful98, Section 2.3]. Hence for Cartier b -divisors determined on X_π ,

$$(18) \quad \langle C_1, \dots, C_n \rangle := C_{1,\pi} \cdots C_{n,\pi}.$$

Boucksom–Favre–Jonsson extend this product to nef Weil b -divisors by monotone approximation; see [BFJ08, Definition 4.3 and Proposition 4.4]. We need only the following finite case.

More explicitly, if $Z_1, \dots, Z_n$ are nef Weil b -divisors, they set

$$(19) \quad \langle Z_1, \dots, Z_n \rangle := \inf_{C_i \geq Z_i} \langle C_1, \dots, C_n \rangle,$$

where the infimum is taken over nef Cartier b -divisors C_i dominating Z_i coefficientwise. Monotonicity of intersection for nef Cartier divisors makes this compatible with (18). The value can in general be $-\infty$, but it is finite in the one-Weil-factor situation used below.

Proposition 3.3 (One Weil factor). *Let Z be a nef Weil b -divisor and $C_2, \dots, C_n$ Cartier b -divisors. Then $\langle Z, C_2, \dots, C_n \rangle$ is finite and multilinear in the Cartier variables. If the C_i are determined on X_π , then*

$$(20) \quad \langle Z, C_2, \dots, C_n \rangle = Z_\pi \cdot C_{2,\pi} \cdots C_{n,\pi}.$$

The same holds when Z is a difference of two nef Weil b -divisors.

Proof. We first define the pairing and then prove the fixed-model formula.

Step 1: extension from nef to signed Cartier factors. By theorem 3.1, every Cartier b -divisor can be written

$$(21) \quad C_i = A_i - B_i$$

with A_i and B_i nef Cartier b -divisors: split an expression (15) into its positive and negative parts. For nef Cartier factors $T_2, \dots, T_n$, the number $\langle Z, T_2, \dots, T_n \rangle$ is finite. This is precisely the case of the finiteness statement after [BFJ08, Proposition 4.9], since at most one factor is non-Cartier. Proposition 4.4 of that paper shows that this finite pairing is additive and positively homogeneous in each nef Cartier variable.

We therefore define

$$(22) \quad \langle Z, C_2, \dots, C_n \rangle := \sum_{\varepsilon_2, \dots, \varepsilon_n \in \{0,1\}} (-1)^{\varepsilon_2 + \dots + \varepsilon_n} \langle Z, T_{2,\varepsilon_2}, \dots, T_{n,\varepsilon_n} \rangle,$$

where $T_{i,0} = A_i$ and $T_{i,1} = B_i$. A multiadditive map on a cancellative commutative semigroup has a unique multilinear extension to the vector space it generates. Hence (22) is independent of all choices in (21). For one signed test variable, this is exactly the extension constructed in [BFJ08, Proposition 4.6]: after fixing nef factors for which the intersection is finite, an arbitrary Cartier divisor is allowed as the test variable. Formula (22) is the simultaneous group completion in all Cartier variables. This proves finiteness and multilinearity without imposing nefness on any C_i .

Step 2: computation on a fixed model. It is enough by multilinearity to suppose that every C_i is nef Cartier. By [BFJ08, Proposition 3.13], there is a decreasing sequence of nef Cartier b -divisors Z_k converging coefficientwise to Z . For each k , choose a common higher model X_{π_k} that dominates X_π and determines Z_k . Computing on X_{π_k} and pushing down to X_π , the ordinary projection formula and [BFJ08, Proposition 4.2] give

$$\langle Z_k, C_2, \dots, C_n \rangle = (Z_k)_\pi \cdot C_{2,\pi} \cdots C_{n,\pi}.$$

The left side decreases to $\langle Z, C_2, \dots, C_n \rangle$ by [BFJ08, Proposition 4.4]. On the right, $(Z_k)_\pi \rightarrow Z_\pi$ in the finite-dimensional space $\text{Div}(\pi)$, so continuity of the ordinary intersection form gives (20). The signed case follows from (22).

Finally, if $Z = Z^+ - Z^-$ is a difference of nef Weil b -divisors, set

$$\langle Z, C_2, \dots, C_n \rangle := \langle Z^+, C_2, \dots, C_n \rangle - \langle Z^-, C_2, \dots, C_n \rangle.$$

This is independent of the presentation: if $Z^+ - Z^- = Y^+ - Y^-$, then $Z^+ + Y^- = Y^+ + Z^-$, and additivity for nef Weil divisors gives the same value. Formula (20) follows by taking the difference of the two fixed-model formulas. $\square$

Remark 3.4 (Mixed multiplicity). For primary ideals,

$$(23) \quad -\langle Z(\mathfrak{a}_1), \dots, Z(\mathfrak{a}_n) \rangle = e(\mathfrak{a}_1, \dots, \mathfrak{a}_n).$$

This is formula (4.3) of [BFJ08], based on [Ram74, Ree84]. It is not logically needed below, but explains why ideal multiplication supplies the polarization data.

3.5. Generalized Lelong numbers as intersections.

Proposition 3.5. *For every psh germ u and every $\mathfrak{m}_o$ -primary ideal $\mathfrak{a}$,*

$$(24) \quad \nu_{\varphi_{\mathfrak{a}}}(u) = -\langle Z(u), Z(\mathfrak{a})^{n-1} \rangle.$$

Proof. Theorem B of [BFJ08] gives, for every tame weight φ ,

$$(25) \quad \nu_{\varphi}(u) = \int_{\mathcal{V}} -\widehat{u} \text{MA}(\widehat{\varphi}),$$

where $\mathcal{V}$ is the normalized valuation space. Proposition 4.6 of [BFJ08] defines the formal Monge–Ampère measure by

$$(26) \quad \int_{\mathcal{V}} \widehat{u} \text{MA}(\widehat{\varphi}) = \langle Z(u), Z(\varphi)^{n-1} \rangle.$$

The measure is well-defined here because $Z(\varphi_a)$ is Cartier; finiteness also follows from [theorem 3.3](#). Substitute $\varphi = \varphi_a$ into [\(25\)](#), then use [\(26\)](#) and [\(17\)](#). This gives [\(24\)](#). The minus sign agrees with the convention in [theorem 2.4](#). $\square$

Remark 3.6. This proposition contains the deep analytic input: Demailly approximation and the comparison between analytic and formal Monge–Ampère masses supplied by [\[BFJ08\]](#). After [\(24\)](#), every intersection is computed on a fixed model. The rest of the proof is finite-dimensional intersection theory together with an elementary polynomial argument.

4. STRICT RELATIVE HODGE SEPARATION

The Hodge input must be strict. A semidefinite statement would not show that a divisor with square zero vanishes. Lemma A.2 of [\[BdFF12\]](#) proves semidefiniteness when the remaining factors are nef. We isolate the additional geometric point that makes the form definite when one fixed ample class is used in every remaining slot.

Lemma 4.1 (A simultaneous surface slice). *Let $\pi : X_\pi \rightarrow (\mathbb{C}^n, o)$ be a smooth projective modification with $n \geq 3$, and let H be an exceptional $\mathbb{Q}$ -divisor whose relative class is π -ample. After replacing H by a positive integral multiple, there are divisors $L_1, \dots, L_{n-2} \in |H|$ such that, for*

$$S := L_1 \cap \dots \cap L_{n-2},$$

the following properties hold in a neighborhood of $\pi^{-1}(o)$:

- (a) *S is a smooth integral surface, is not contained in $\text{Exc}(\pi)$, and meets every exceptional stratum properly;*
- (b) *if $Y := \pi(S)$, then $S \rightarrow Y$ factors through a proper birational morphism $S \rightarrow Y^\nu$ to the normalization of the surface germ Y ;*
- (c) *for every exceptional prime E , the restriction $E|_S$ is a nonzero exceptional curve, and $E|_S$ and $F|_S$ have no common curve component whenever $E \neq F$.*

Proof. By [theorem 2.1](#) and homogeneity, we may multiply H and suppose that $\mathcal{O}_{X_\pi}(H)$ is π -very ample. Shrink the representative U of the base germ so that it is affine. Since the fiber $\pi^{-1}(o)$ is proper, there is a finite-dimensional subspace

$$V \subset H^0(\pi^{-1}(U), \mathcal{O}_{X_\pi}(H))$$

that generates $\mathcal{O}_{X_\pi}(H)$ and whose associated morphism is an immersion on a neighborhood of $\pi^{-1}(o)$. To obtain such a V , start with finitely many sections defining a relative embedding into $\mathbb{P}_U^N$ and enlarge their span by multiplying them with pullbacks of local coordinates on U .

Take the L_i to be general members of the base-point-free linear system $\mathbb{P}(V)$. Bertini’s smoothness and irreducibility theorems, applied successively, make their complete intersection smooth and integral near the proper fiber; for the smoothness statement see [\[Har77, Chapter III, Corollary 10.9\]](#). There

are only finitely many exceptional strata, namely the irreducible components of intersections of exceptional primes, so the same general choice makes S meet each of them properly. It also makes S meet the complement of the exceptional locus. This proves (a).

Outside $\pi^{-1}(o)$ the map π is an isomorphism. Endow $Y = \pi(S)$ with its reduced structure. Then Y has dimension two and $S \rightarrow Y$ is an isomorphism away from o . Since S is normal, the universal property of normalization yields a factorization $S \rightarrow Y^\nu \rightarrow Y$; the first map is proper and birational. This proves (b).

Fix an exceptional prime E . The restriction $H|_E$ is ample because E is contained in a fiber of π . Therefore the intersection on the $(n-1)$ -dimensional proper variety E of $n-2$ general members of $|H|_E|$ is a nonzero one-cycle. This is exactly

$$E|_S = E \cdot L_1 \cdots L_{n-2} = E \cdot H^{n-2},$$

and it is contracted to o . If $E \neq F$, then $\dim(E \cap F) \leq n-2$. Proper intersection with the $n-2$ general hyperplanes gives

$$\dim(E \cap F \cap S) \leq 0.$$

Thus $E|_S$ and $F|_S$ share no curve component. This proves (c). $\square$

Lemma 4.2 (Strict relative Hodge index). *Let $\pi : X_\pi \rightarrow (\mathbb{C}^n, o)$ be a smooth projective modification, $n \geq 2$. Let $H \in \text{Div}(\pi)_\mathbb{Q}$ have a π -ample relative numerical class. Then*

$$(27) \quad Q_H(D_1, D_2) := D_1 \cdot D_2 \cdot H^{n-2}$$

is negative definite on $\text{Div}(\pi)$. Thus

$$(28) \quad D \neq 0 \implies D^2 \cdot H^{n-2} < 0.$$

Proof. If $n = 2$, the intersection matrix of the exceptional curves of a proper birational morphism from a smooth surface to a normal surface germ is negative definite. This is Mumford's negativity theorem [Mum61]. Hence $D^2 < 0$ for every nonzero exceptional real divisor D .

Assume $n \geq 3$. Multiplying H by a positive integer changes Q_H by a positive scalar, so we may apply theorem 4.1. Write $D = \sum_E d_E E$ and choose E_0 with $d_{E_0} \neq 0$. By theorem 4.1(c), $E_0|_S$ has a curve component that is not a component of $F|_S$ for any $F \neq E_0$. Its coefficient in $D|_S$ is therefore nonzero. Thus

$$(29) \quad D \neq 0 \implies D|_S \neq 0.$$

All components of $D|_S$ are exceptional for the proper birational surface map $S \rightarrow Y^\nu$. Mumford's negativity theorem gives $(D|_S)^2 < 0$ whenever $D \neq 0$. Finally, associativity of local intersection products gives

$$(D|_S)^2 = D^2 \cdot L_1 \cdots L_{n-2} = D^2 \cdot H^{n-2}.$$

This proves (28) and the lemma. $\square$

Remark 4.3 (Nef versus ample). Lemma A.2 of [BdFF12] gives negative semidefiniteness for merely nef remaining factors. Semidefiniteness permits nonzero null vectors. Relative ampleness ensures that every exceptional prime cuts a nonzero curve and makes (29) available, yielding strict negativity.

Remark 4.4 (Existence of H). Such an H exists by theorem 2.1. The point of theorem 4.2 is not existence but strictness: the same ample class cuts every exceptional prime in a nonzero curve, so no nonzero exceptional divisor survives as a null vector.

Example 4.5 (Sign check on the blowup of the origin). Let $\pi : \text{Bl}_o \mathbb{C}^n \rightarrow \mathbb{C}^n$ and let $E \simeq \mathbb{P}^{n-1}$ be the exceptional divisor. Then $H = -E$ is π -ample and

$$E^2 \cdot H^{n-2} = (-1)^{n-2} E^n = -1,$$

since $E^n = (-1)^{n-1}$. This agrees with the sign convention in theorem 2.4 and with theorem 4.2.

5. POLARIZATION OF DIAGONAL IDENTITIES

Lemma 5.1 (Polynomial vanishing on a positive lattice). *If a polynomial P in r real variables vanishes at every point of $\mathbb{N}_{>0}^r$, then $P = 0$.*

Proof. Induct on r . For $r = 1$, a nonzero polynomial has finitely many roots. Write $P = \sum_{j=0}^d P_j(x_1, \dots, x_{r-1})x_r^j$. After fixing positive integers $k_1, \dots, k_{r-1}$, the resulting polynomial in x_r has infinitely many roots, so every coefficient $P_j(k_1, \dots, k_{r-1})$ is zero. Induction gives $P_j = 0$ for every j . $\square$

Lemma 5.2 (Polarization by products of ideals). *Assume $n \geq 2$. Let W be a difference of nef Weil b -divisors. If*

$$(30) \quad \langle W, Z(\mathfrak{a})^{n-1} \rangle = 0$$

for every $\mathfrak{m}_o$ -primary ideal $\mathfrak{a}$, then

$$(31) \quad \langle W, Z(\mathfrak{a}_1), \dots, Z(\mathfrak{a}_{n-1}) \rangle = 0$$

for all primary ideals $\mathfrak{a}_i$.

Proof. Set $r = n - 1$ and $A_i = Z(\mathfrak{a}_i)$. For $\mathbf{k} = (k_1, \dots, k_r) \in \mathbb{N}_{>0}^r$, the ideal $\mathfrak{a}(\mathbf{k}) = \mathfrak{a}_1^{k_1} \cdots \mathfrak{a}_r^{k_r}$ is primary and (14) gives $Z(\mathfrak{a}(\mathbf{k})) = \sum_i k_i A_i$. Hence

$$(32) \quad P(\mathbf{k}) := \langle W, (k_1 A_1 + \cdots + k_r A_r)^r \rangle = 0.$$

By multilinearity,

$$(33) \quad P(\mathbf{k}) = \sum_{\alpha_1 + \cdots + \alpha_r = r} \frac{r!}{\alpha_1! \cdots \alpha_r!} k_1^{\alpha_1} \cdots k_r^{\alpha_r} \langle W, A_1^{\alpha_1}, \dots, A_r^{\alpha_r} \rangle.$$

By theorem 5.1, this polynomial is identically zero. The coefficient of $k_1 \cdots k_r$ is $r! \langle W, A_1, \dots, A_r \rangle$, so that mixed intersection vanishes. $\square$

Example 5.3 (Dimension three). For $n = 3$, put $A = Z(\mathfrak{a})$ and $B = Z(\mathfrak{b})$. The diagonal identities for A , B , and $A + B = Z(\mathfrak{ab})$ give

$$0 = \langle W, (A + B)^2 \rangle = \langle W, A^2 \rangle + 2\langle W, A, B \rangle + \langle W, B^2 \rangle.$$

The first and last terms vanish, so $\langle W, A, B \rangle = 0$. The general argument is the degree- $(n - 1)$ multinomial version of this calculation.

Proposition 5.4 (Extension to all Cartier tests). *Assume $n \geq 2$. Under the hypothesis of [theorem 5.2](#),*

$$(34) \quad \langle W, C_1, \dots, C_{n-1} \rangle = 0$$

for arbitrary Cartier b -divisors C_i .

Proof. Use [theorem 3.1](#) to write each C_i as a finite real linear combination of divisors $Z(\mathfrak{a}_{ij})$. Multilinearity expands the left side of (34) into finitely many mixed ideal intersections. Every term vanishes by [theorem 5.2](#). $\square$

6. PROOF OF THE MAIN THEOREM

Proof of [theorem 1.1](#). We prove the implications separately.

Step 1: (i) $\Rightarrow$ (ii). By (10), condition (i) is equivalent to $Z(u) = Z(v)$, hence to $\hat{u} = \hat{v}$. Theorem B of [BFJ08] expresses generalized Lelong numbers with tame weights by (25). Equal valuative transforms have equal integrals.

Step 2: (ii) $\Rightarrow$ (iii). The weight $\varphi_{\mathfrak{a}}$ is tame by [BFJ08, Lemma 5.10], as explained after (16); hence it may be substituted into condition (ii).

Step 3: obtain diagonal intersection identities. If $n = 1$, then $(dd^c \varphi)^0 = 1$ and $\nu_{\varphi}(u) = (dd^c u)(\{o\}) = \nu^L(u, o)$. A proper modification of a smooth curve germ is an isomorphism, so the conclusion follows. Assume $n \geq 2$ and set

$$(35) \quad W := Z(u) - Z(v).$$

It is a difference of nef Weil b -divisors by [theorem 2.3](#). Using condition (iii) and [theorem 3.5](#),

$$\begin{aligned} 0 &= \nu_{\varphi_{\mathfrak{a}}}(u) - \nu_{\varphi_{\mathfrak{a}}}(v) \\ &= -\langle Z(u), Z(\mathfrak{a})^{n-1} \rangle + \langle Z(v), Z(\mathfrak{a})^{n-1} \rangle \\ &= -\langle W, Z(\mathfrak{a})^{n-1} \rangle. \end{aligned}$$

Therefore

$$(36) \quad \langle W, Z(\mathfrak{a})^{n-1} \rangle = 0 \quad \text{for every primary ideal } \mathfrak{a}.$$

Step 4: pass from diagonal to arbitrary Cartier tests. Lemma 5.2 turns (36) into all mixed ideal identities. Proposition 5.4 then gives

$$(37) \quad \langle W, C_1, \dots, C_{n-1} \rangle = 0$$

for all Cartier b -divisors C_i .

Step 5: separate W on a fixed model. Fix an arbitrary smooth projective model X_{π} . Its trace $W_{\pi} = Z(u)_{\pi} - Z(v)_{\pi}$ is a real exceptional divisor. It

defines the Cartier b -divisor $\overline{W}_\pi$ determined on X_π . Choose a relatively ample exceptional rational divisor H as in [theorem 4.4](#), also viewed as a Cartier b -divisor.

In [\(37\)](#), take

$$C_1 = \overline{W}_\pi, \quad C_2 = \cdots = C_{n-1} = H.$$

There are $n - 2$ copies of H ; for $n = 2$ there are none. We obtain

$$\langle W, \overline{W}_\pi, H^{n-2} \rangle = 0.$$

All Cartier factors are determined on X_π , so the projection formula [\(20\)](#) gives

$$(38) \quad W_\pi^2 \cdot H^{n-2} = 0.$$

By [theorem 4.2](#), a nonzero exceptional divisor has strictly negative square against H^{n-2} . Hence $W_\pi = 0$.

Step 6: return to infinitely near points. The model was arbitrary, so every incarnation of W is zero. Thus $W = 0$ and $Z(u) = Z(v)$. Equation [\(10\)](#) gives equality of all infinitely near Lelong numbers, which is condition (i). $\square$

Remark 6.1 (The surface case). When $n = 2$, [\(36\)](#) is already linear in $Z(\mathfrak{a})$. Since ideal divisors span all Cartier b -divisors, it gives $\langle W, C \rangle = 0$ for all Cartier C . Taking $C = \overline{W}_\pi$ and using Mumford negativity gives $W_\pi = 0$. Polarization is needed only in dimensions at least three.

Remark 6.2 (First-variation interpretation). For ideal singularities in every entry, [\(23\)](#) identifies the intersections with mixed multiplicities. The diagonal functional

$$A \longmapsto -\langle Z(u), A^{n-1} \rangle$$

determines all mixed values by polarization. Relative Hodge separation then recovers $Z(u)$. This explains why fundamental solutions of the formal Monge–Ampère operator are unnecessary here: individual valuations need not be isolated by positive measures once signed Cartier tests are available.

Remark 6.3 (Logical dependencies). The non-elementary inputs are:

- (1) nefness of $Z(u)$, [\[BFJ08, Theorem 5.1\]](#);
- (2) generation of Cartier b -divisors by ideal divisors, [\[BFJ08, Proposition 1.14\]](#);
- (3) decreasing Cartier approximation, [\[BFJ08, Proposition 3.13\]](#), and the intersection results [\[BFJ08, Propositions 4.2, 4.4, 4.6, and 4.9\]](#);
- (4) the generalized Lelong formula, [\[BFJ08, Theorem B\]](#); and
- (5) surface negativity from [\[Mum61\]](#), together with the Bertini slicing used in [theorem 4.1](#).

The polynomial argument and the strict separation step are proved here.

REFERENCES

- [BdFF12] S. Boucksom, T. de Fernex, and C. Favre, *The volume of an isolated singularity*, Duke Math. J. **161** (2012), no. 8, 1455–1520.
- [BFJ08] S. Boucksom, C. Favre, and M. Jonsson, *Valuations and plurisubharmonic singularities*, Publ. Res. Inst. Math. Sci. **44** (2008), no. 2, 449–494.
- [Dem87] J.-P. Demailly, *Nombres de Lelong généralisés, théorèmes d'intégralité et d'analyticité*, Acta Math. **159** (1987), no. 3–4, 153–169.
- [DF22] N.-B. Dang and C. Favre, *Intersection theory of nef b -divisor classes*, Compos. Math. **158** (2022), no. 7, 1563–1594.
- [FJ05] C. Favre and M. Jonsson, *Valuative analysis of planar plurisubharmonic functions*, Invent. Math. **162** (2005), no. 2, 271–311.
- [Ful98] W. Fulton, *Intersection theory*, second ed., Ergebnisse der Mathematik und ihrer Grenzgebiete, vol. 2, Springer-Verlag, Berlin, 1998.
- [Har77] R. Hartshorne, *Algebraic geometry*, Graduate Texts in Mathematics, vol. 52, Springer-Verlag, New York–Heidelberg, 1977.
- [GZ15] Q. Guan and X. Zhou, *A proof of Demailly's strong openness conjecture*, Ann. of Math. (2) **182** (2015), no. 2, 605–616.
- [Laz04] R. Lazarsfeld, *Positivity in algebraic geometry I*, Ergebnisse der Mathematik und ihrer Grenzgebiete, vol. 48, Springer-Verlag, Berlin, 2004.
- [Mum61] D. Mumford, *The topology of normal singularities of an algebraic surface and a criterion for simplicity*, Publ. Math. Inst. Hautes Études Sci. **9** (1961), 5–22.
- [Ram74] C. P. Ramanujam, *On a geometric interpretation of multiplicity*, Invent. Math. **22** (1973/74), 63–67.
- [Ree84] D. Rees, *Generalizations of reductions and mixed multiplicities*, J. London Math. Soc. (2) **29** (1984), no. 3, 397–414.
- [Sho03] V. V. Shokurov, *Prelimiting flips*, Proc. Steklov Inst. Math. **240** (2003), no. 1, 75–213.

DECLARATION ON THE USE OF ARTIFICIAL INTELLIGENCE

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